SVM Image Classification: Apple vs Banana

Step-by-step numerical handout using handcrafted colour feature vectors

Image → Feature Extraction → Feature Vector → SVM → Class Label

1. Handcrafted Feature Vector

For a simple classroom example, represent each image using the mean Red, Green and Blue intensities. For an image with N pixels:

$$R_mean = (1/N)\,\Sigma\, R_p \qquad G_mean = (1/N)\,\Sigma\, G_p \qquad B_mean = (1/N)\,\Sigma\, B_p$$

$$x = [R_mean,\ G_mean,\ B_mean]$$

This converts an image containing thousands of pixel values into a compact 3-dimensional numerical feature vector.

2. Training Data

Training image	Feature vector x = [R, G, B]	Class label y
Apple 1	[210, 70, 55]	+1
Apple 2	[205, 65, 50]	+1
Banana 1	[225, 205, 60]	−1
Banana 2	[230, 210, 70]	−1

We choose Apple = +1 and Banana = −1. In this toy dataset, the Green-channel feature already separates the two classes clearly:

$$\text{Apple: } G = 65,\ 70 \qquad \text{Banana: } G = 205,\ 210$$

3. Identify the Closest Training Samples

The nearest Apple to the Banana class has G = 70. The nearest Banana to the Apple class has G = 205. These are the support-vector locations in this simplified one-dimensional separation.

$$G_A = 70 \qquad G_B = 205$$

The maximum-margin decision boundary lies midway between the two support vectors:

$$G_boundary = (70 + 205)/2 = 137.5$$

Therefore: G < 137.5 → Apple | G > 137.5 → Banana

4. Linear SVM Equation

For a linear binary SVM:

$$f(x) = w^T x + b$$

where w contains the feature weights and b is the bias. Here:

$$x = [R, G, B]^T \qquad w = [w_R, w_G, w_B]^T$$

Because this simplified dataset can be separated using only G, choose w_R = 0 and w_B = 0. We solve for w_G and b from the support-vector conditions.

5. Solve w and b Step by Step

For a hard-margin SVM, support vectors satisfy $y(w^T x + b) = 1$. Therefore:

$$\text{Apple support vector } (G = 70): \quad 70w_G + b = +1 \quad \ldots(1)$$

$$\text{Banana support vector } (G = 205): \quad 205w_G + b = -1 \quad \ldots(2)$$

Subtract Eq. (1) from Eq. (2):

$$205w_G - 70w_G = -1 - 1$$

$$135w_G = -2$$

$$w_G = -2/135 \approx -0.014815$$

Substitute into Eq. (1):

$$(-0.014815)(70) + b = 1$$

$$-1.037 + b = 1$$

$$b \approx 2.037$$

Final model: w = [0, −0.014815, 0] b = 2.037

6. Final Decision Function

$$f(x) = 0R - 0.014815G + 0B + 2.037$$

$$f(x) = 2.037 - 0.014815G$$

Prediction rule: f(x) > 0 → Apple | f(x) < 0 → Banana

7. Test a New Apple Image

Suppose the new image has feature vector:

$$x = [208, 68, 52]$$

Use the learned SVM decision function:

$$f(x) = 2.037 - 0.014815(68)$$

$$f(x) = 2.037 - 1.0074 \approx 1.030$$

Since f(x) > 0, the predicted class is APPLE.

8. Test a New Banana Image

Suppose:

$$x = [228, 208, 65]$$

$$f(x) = 2.037 - 0.014815(208)$$

$$f(x) = 2.037 - 3.0815 \approx -1.045$$

Since f(x) < 0, the predicted class is BANANA.

9. Verify the Decision Boundary

At the decision boundary, f(x) = 0:

$$2.037 - 0.014815G = 0$$

$$G \approx 137.5$$

This agrees exactly with the midpoint calculated earlier.

10. Margin Calculation

Distance from the boundary to each support vector:

$$137.5 - 70 = 67.5 \quad \text{and} \quad 205 - 137.5 = 67.5$$

The full margin between the two supporting hyperplanes is:

$$\text{Margin} = 2 \,/\, ||w||$$

$$||w|| = 0.014815 \quad \Rightarrow \quad \text{Margin} \approx 2/0.014815 \approx 135$$

This equals 205 − 70 = 135, confirming the geometry.

11. Generalization to Real Image Classification

Real image classification normally uses richer handcrafted features. A complete feature vector may contain colour, texture and shape descriptors:

$$x = [R_mean, G_mean, B_mean, Contrast, Energy, Homogeneity, Correlation, HOG_1, ..., HOG_n]$$

The SVM then learns one weight for each feature:

$$f(x) = w_1x_1 + w_2x_2 + ... + w_dx_d + b$$

In software, the SVM optimization automatically identifies the support vectors and learns w and b.

12. Summary

Step	Meaning
1	Extract handcrafted features from each image
2	Form a numerical feature vector x
3	Assign class labels (+1 / −1)
4	SVM finds the maximum-margin separating boundary
5	Learn w and b
6	For a new image compute $f(x) = w^Tx + b$
7	Use the sign of f(x) to assign the class

Key idea: SVM classifies feature vectors by finding a separating hyperplane with the maximum possible margin between classes.

Teaching note: This numerical example is intentionally simplified so every calculation can be verified manually.